Department of Electrical and Computer Engineering
COMSATS University Islamabad, Attock Campus

Lab Rubrics Form		
Lab Name	Ethernet Cabling and Twisted Pair Cable Construction	
Lab Number	01	
Submitted by:		
Student Names		**Registration #**
Sumaira Safeer		FA22-BCE-019

Rubrics		**Marks**		
		1	**2**	**3**
PLO-5 Modern Tools Usage	Ability to execute the experiment utilizing hardware, software tools, or a combination of both.			
PLO-10 Communication Skills	Present lab activities effectively through detailed lab reports, oral examinations, and engaging presentations.			

Lab Tasks

Post Lab Task 1:
Objective:
This lab exercise is designed to give students hands on experience of building CAT-5 UTP Ethernet patch cables.

Answer the following:
I. What is the advantage of having twists in a twisted pair cable?

- Twisting the pairs helps to cancel out electromagnetic interference (EMI) and crosstalk from adjacent pairs.
- It reduces noise and signal degradation, improving data transmission quality.
- Twisting ensures self-shielding, preventing external interference from affecting signal integrity.

II. Identify the differences between twisted pair and coaxial cables?

Feature:	**Twisted Pair Cable:**	**Coaxial Cable:**
Structure	Consists of pairs of twisted copper wires.	Single copper core with insulation and metallic shielding.
Shielding	UTP has no shielding; STP has shielding.	Has better shielding against interference.
Interference Resistance	Moderate (UTP), High (STP).	Very high interference resistance.
Speed	10 Mbps - 1 Gbps (UTP).	10 Mbps - 100 Mbps.
Cost	Less expensive.	More expensive.
Maximum Length	100 meters.	500 meters.

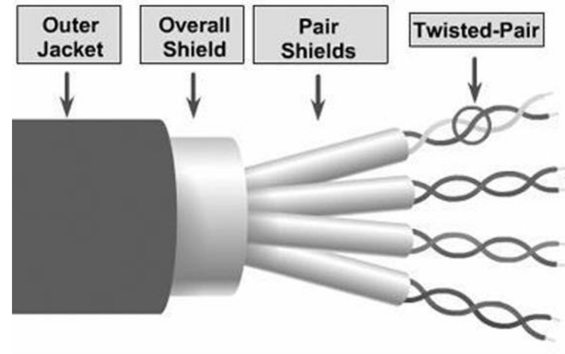

Figure 1-3 Shielded Twisted-Pair Cable

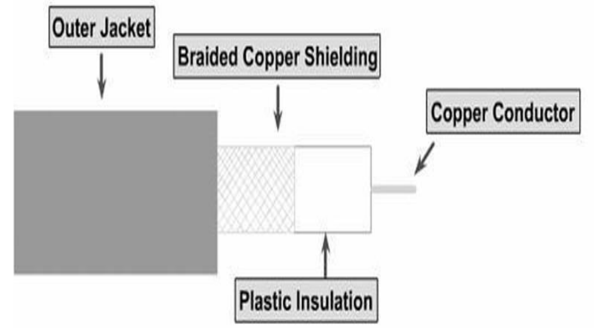

Figure 1-4 Coaxial Cable

III. Which type of twisted pair cable will you use to connect?

- **Router to Hub:** Straight-through cable.
- **Router to Router:** Crossover cable.
- **PC to Hub:** Straight-through cable.
- **PC to PC:** Crossover cable.

IV. Can you connect a hub to another hub or a switch to another switch using a straight-through cable? Explain your answer.

- No, a straight-through cable cannot be used for connecting a hub to another hub or a switch to another switch.
- Hubs and switches operate at the same layer (Layer 2), and to establish proper communication, a crossover cable is required.
- However, modern switches with auto-MDI/MDIX capability can automatically detect the cable type and adjust accordingly.

V. Summarize in tabular form the features of UTP, STP, and Coaxial cables in terms of speed, average cost per node, connector size, and maximum cable length.

Feature:	**UTP (Unshielded Twisted Pair):**	**STP (Shielded Twisted Pair):**	**Coaxial Cable:**
Speed	10 Mbps - 1Gbps.	10 Mbps - 100 Mbps.	10Mbps - 100 Mbps.
Average Cost per Node	Least expensive.	Moderately expensive.	Inexpensive.
Connector Size	Small (RJ45).	Medium to large (STP connector /RJ45).	Medium (BNC).

Maximum Cable Length	100 meters.	100 meters.	500 meters.

VI. What is the difference between Thicknet and Thinnet? Write the connector names of Thicknet and Thinnet.

Difference Between Thicknet and Thinnet:

Thicknet and Thinnet are both types of coaxial cables used in early Ethernet networks, but they differ in several key aspects. Thicknet, also known as **10BASE5**, is a thicker and more rigid cable, approximately 1 cm in diameter. It can support cable lengths of up to **500 meters**, making it suitable for use as a backbone cable in large networks. However, its rigid structure makes installation difficult and expensive. Thicknet uses **vampire taps** and AUI (Attachment Unit Interface) connectors to attach devices, which adds to the complexity of setup.

On the other hand, Thinnet, known as **10BASE2**, is a thinner and more flexible coaxial cable with a diameter of about **0.35 cm**.Both Thicknet and Thinnet have been largely replaced by **twisted-pair cables (UTP)**, which offer even greater flexibility, easier installation, and better performance in modern Ethernet networks.

Thicknet (10BASE5):

- Has a thicker diameter (1 cm).
- Used for backbone networks with longer transmission distances.
- Requires a vampire tap for connection.

Thinnet(10BASE2):

- Has a thinner diameter (0.35 cm).
- More flexible and easier to install.
- Uses BNC connectors for connections.

Connectors:

- **Thicknet:** AUI (Attachment Unit Interface) and Vampire Tap.
- **Thinnet:** BNC (Bayonet Neill-Concelman).